



GATE CSE — Computer Networks

Complete Numericals & Formulas Revision Guide



Formulas •



Examples •



Traps •



Key Points

🎯 1. Transmission & Propagation Delay (VERY FREQUENT)

Two key delay types you MUST distinguish clearly:

Transmission Delay	Propagation Delay
Depends on packet size (L) & bandwidth (B)	Depends on distance (d) & signal speed (v)
Formula: $T_t = L / B$	Formula: $T_p = d / v$
Time to push ALL bits onto wire	Time for a bit to travel from A → B
Example: 1000 bits @ 1 Mbps = 1 ms	Example: 100 km @ 2×10^8 m/s = 0.5 ms



Key Formulas

T_t	Transmission Delay = L (bits) / B (bps)
T_p	Propagation Delay = d (meters) / v (m/s)
Total	Total Delay = $T_t + T_p$ (for single link)



Example: When does first bit arrive vs last bit?

Given	Packet = 2000 bits, BW = 1 Mbps, Distance = 300 km, $v = 3 \times 10^8$ m/s
T_t	$2000 / 1 \times 10^6 = 2$ ms
T_p	$300,000 / 3 \times 10^8 = 1$ ms
First bit arrives at	$T_p = 1$ ms (it starts propagating immediately!)
Last bit arrives at	$T_t + T_p = 2 + 1 = 3$ ms



Common Traps / Tricks

✗ First bit sent ≠ First bit received. First bit arrives after ONLY T_p (not $T_t + T_p$)
✗ Last bit arrives = $T_t + T_p$ (both delays add up)
✗ Do NOT confuse L (packet size in bits) with d (distance in meters)
✗ Bandwidth-delay product ≠ transmission delay — they measure different things

© 2. Bandwidth-Delay Product (BDP)

BDP tells you how many bits are 'in flight' (in the pipe) at any given time.

BDP	$\text{BDP} = \text{Bandwidth (bps)} \times \text{Propagation Delay (s)} = \text{bits}$
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✓ BDP = maximum data that can be in transit simultaneously on a link
✓ Required sender buffer size = BDP (for full pipe utilization)
✓ Window size (in bits) must be $\geq$ BDP for efficient transmission

🔗 Example: Buffer / Window Size

Link Speed	100 Mbps
RTT (Prop. Delay)	20 ms
BDP	$100 \times 10^6 \times 0.02 = 2,000,000 \text{ bits} = 2 \text{ Mb}$
Meaning	Sender can have 2 Mb 'in the air' — window must be $\geq 2 \text{ Mb}$

© 3. Sliding Window Protocol (Super Important)

Protocol Comparison

Protocol	Window Size	Seq Bits (n)	Max Efficiency
Stop & Wait	1	1 bit	$1 / (1 + 2a)$
Go-Back-N (GBN)	$2^n - 1$	n bits	$W / (1 + 2a)$ if $W < 1 + 2a$
Selective Repeat	2^{n-1}	n bits	$W / (1 + 2a)$ if $W < 1 + 2a$

📐 Formulas

a	$a = T_p / T_t \quad (\text{propagation-to-transmission ratio})$
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S&W Eff	$\text{Efficiency} = 1 / (1 + 2a)$
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GBN Wmax	$\text{Max Window for GBN} = 2^n - 1$
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SR Wmax	$\text{Max Window for SR} = 2^{n-1}$
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Efficiency	$\text{If } W \geq 1 + 2a \rightarrow \text{Efficiency} = 1 \text{ (100\%)}$
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Example: GATE-style: Find efficiency of GBN (n=3 bits, Tp=20ms, Tt=5ms)

n = 3 bits	Window size $W_{\max} = 2^3 - 1 = 7$
a = T_p/T_t	$20 / 5 = 4$
1 + 2a	$1 + 8 = 9$
W vs 1+2a	$7 < 9$, so use $W/(1+2a)$
Efficiency	$7 / 9 = 77.8\%$

✓ GBN: On error, retransmit the erroneous frame AND all subsequent frames
✓ SR: Retransmit ONLY the erroneous frame (uses receiver buffer)
✓ SR needs receiver buffer; GBN does not
✓ Sequence numbers for GBN: $2^n - 1$ windows SR: 2^{n-1} windows

4. CRC — Error Detection

CRC uses binary polynomial division (mod-2 arithmetic). No carry, no borrow — XOR only!

Step-by-Step CRC Process

Step 1	Append (r) zeros to the message, where r = degree of generator polynomial
Step 2	Divide appended message by generator using mod-2 division (XOR)
Step 3	Remainder = CRC bits (r bits)
Step 4	Transmitted frame = Original message + CRC remainder
Verify	Divide received frame by generator — if remainder = 0, no error detected

Example: CRC Calculation

Message (M)	1101011011
Generator (G)	10011 (degree = 4, so append 4 zeros)
Appended M	11010110110000
Divide by G	XOR-based long division...
Remainder (CRC)	1110
Transmitted Frame	1101011011 1110

Hamming Code Basics

Hamming	$2^r \geq m + r + 1$ (find min parity bits r for m data bits)
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✓ Hamming distance = minimum number of bit flips to go from one valid codeword to another
✓ To detect d errors: need Hamming distance $\geq d + 1$
✓ To correct d errors: need Hamming distance $\geq 2d + 1$
✓ Parity bits at positions: 1, 2, 4, 8, 16, ... (powers of 2)

🎯 5. Subnetting & IP Addressing

Key Formulas

Hosts	Usable Hosts per subnet = $2^h - 2$ (h = host bits)
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Subnets	Number of subnets = 2^s (s = borrowed subnet bits)
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CIDR	IP/prefix e.g. 192.168.1.0/24 → 24 network bits, 8 host bits
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🔗 Example: Subnet a /24 into 8 equal subnets

Original	192.168.10.0/24 (256 addresses)
Need 8 subnets	Borrow 3 bits ($2^3 = 8$)
New prefix	/27 ($24 + 3 = 27$)
Hosts per subnet	$2^5 - 2 = 30$ usable hosts
Subnet 1	192.168.10.0/27 → .1 to .30 (broadcast: .31)
Subnet 2	192.168.10.32/27 → .33 to .62 (broadcast: .63)

✓ Network ID: All host bits = 0
✓ Broadcast ID: All host bits = 1
✓ Valid host range: Network ID + 1 to Broadcast - 1
✓ Subtract 2 for network & broadcast addresses when counting usable hosts

⚠ Common Traps / Tricks

✗ /32 = single host, /31 = point-to-point (0 usable traditionally)
✗ Do NOT subtract 2 for counting total addresses — only for usable hosts

✗ Class A default mask: /8, Class B: /16, Class C: /24

6. Routing Algorithms

Dijkstra (Link State)	Bellman-Ford (Distance Vector)
Centralized: each router knows full topology	Distributed: routers exchange with neighbors only
$O(n^2)$ time complexity	$O(VE)$ time complexity
Converges faster; no routing loops	Can have count-to-infinity problem
Used in OSPF	Used in RIP
Greedy: always expand minimum cost node	Iterates until no distance updates occur

Dijkstra Step-by-Step Template

1. Initialize: source = 0, all others = ∞ . Mark source as visited.
2. For current node, update all unvisited neighbors: $d[v] = \min(d[v], d[u] + w(u,v))$
3. Pick the unvisited node with smallest $d[]$ value -> mark as visited.
4. Repeat until all nodes visited.

7. Ethernet / CSMA-CD Numericals

Minimum frame size ensures the sender is STILL transmitting when a collision signal returns.

MinFrame	Min Frame Size = $2 \times T_p \times \text{Bandwidth}$ = $2 \times \text{BDP}$
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64 bytes	Ethernet: $2 \times T_p \times B \geq 512 \text{ bits (64 bytes)}$ -> max segment: 2500m
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🔧 Example: Why minimum 64 bytes?

Max round-trip time	$2 \times T_p$ (worst case: collision at far end)
Condition	$T_t \geq 2 \times T_p$ -> frame must take $\geq$ round-trip time to transmit
Ethernet max T_p	$\sim 25.6 \mu\text{s}$ -> $2 \times T_p = 51.2 \mu\text{s}$
@ 10 Mbps	$10 \times 10^6 \times 51.2 \times 10^{-6} = 512 \text{ bits} = 64 \text{ bytes}$

✓ If frame < min size, sender finishes before collision signal returns -> collision undetected

✓ Collision domain must be bounded so $2 \times T_p \leq T_t$

✓ Repeaters extend collision domain; switches/routers break it

8. ALOHA Protocols (Very Common)

Pure ALOHA	Slotted ALOHA
Transmit anytime	Transmit only at slot boundaries
$S = G \times e^{(-2G)}$	$S = G \times e^{(-G)}$
Max $G = 0.5 \rightarrow$ Max $S = 1/(2e) \approx 18.4\%$	Max $G = 1 \rightarrow$ Max $S = 1/e \approx 36.8\%$
Vulnerable period = $2 \times$ frame time	Vulnerable period = $1 \times$ frame time

Pure	$S = G \cdot e^{(-2G)}$ [max efficiency $\approx 18.4\%$ at $G = 0.5$]
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Slotted	$S = G \cdot e^{(-G)}$ [max efficiency $\approx 36.8\%$ at $G = 1.0$]
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📌 Example: Find S when $G = 0.5$ (Slotted ALOHA)

$G = 0.5$	$S = 0.5 \times e^{(-0.5)}$
$e^{(-0.5)} \approx$	0.6065
$S =$	$0.5 \times 0.6065 = 0.3033 = 30.3\%$

✓ Slotted ALOHA is EXACTLY 2× more efficient than Pure ALOHA
✓ G = offered load (frames per frame-time); S = throughput (successful frames)
✓ At max efficiency: for Pure ALOHA $G=0.5$, for Slotted $G=1$

🎯 9. Leaky Bucket & Token Bucket

Leaky Bucket	Token Bucket
Fixed output rate (constant)	Variable output rate (bursty allowed)
Excess packets DROPPED	Burst allowed up to token count
No burst allowed	Max burst = bucket size (b)
Output = $\min(\text{arrival}, r)$	Max output rate = $r + b/T$

Token Bucket	Max burst output = $(\text{token_rate} \times \text{time}) + \text{bucket_capacity}$
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📌 Example: Token Bucket: How long can max burst last?

Max line rate (M)	10 Mbps
Token rate (r)	2 Mbps
Bucket size (b)	1 Mb
Burst duration	$b / (M - r) = 1 \text{ Mb} / (10 - 2) \text{ Mbps} = 0.125 \text{ s} = 125 \text{ ms}$

🎯 10. Packet Switching — End-to-End Delay

Store-and-forward: router must receive ENTIRE packet before forwarding it.

E2E

$$\text{Total} = (n+1) \times T_t + n \times T_p \quad [n \text{ routers, ignoring queuing/processing}]$$

📌 Example: 3 routers, 4 links, packet = 1000 bits, BW = 1 Mbps, link delay = 1ms

Tt per link	$1000 / 10^6 = 1 \text{ ms}$
Links = 4	Transmission delay = $4 \times 1 \text{ ms} = 4 \text{ ms}$
Propagation (3 routers)	$3 \times 1 \text{ ms} = 3 \text{ ms}$ (one per link, assuming 4 links)
Total delay	$4 \text{ ms} + 3 \text{ ms} = 7 \text{ ms}$ (ignoring queuing)

Delay Type	Description
Transmission	Time to push all bits onto wire = L/B
Propagation	Time for signal to travel link = d/v
Queuing	Wait time in router queue (varies with load)
Processing	Router looks up route, error check (usually given or negligible)

🎯 11. Stop-and-Wait Throughput

Throughput

$$\text{Throughput} = \text{Bandwidth} \times \text{Efficiency} = B / (1 + 2a)$$

a

$$a = T_p / T_t$$

📌 Example: S&W: BW=1Mbps, frame=1000bits, d=150km, v=3×10⁸m/s

Tt	$1000 / 10^6 = 1 \text{ ms}$
Tp	$150,000 / 3 \times 10^8 = 0.5 \text{ ms}$
a = Tp/Tt	$0.5 / 1 = 0.5$
Efficiency	$1 / (1 + 2 \times 0.5) = 1/2 = 50\%$
Throughput	$1 \text{ Mbps} \times 50\% = 500 \text{ Kbps}$

🎯 12. TCP Throughput (Advanced)

TCP

$$\text{Throughput} \approx \text{MSS} / (\text{RTT} \times \sqrt{p})$$

Where: MSS = Max Segment Size, RTT = Round Trip Time, p = packet loss probability

✓ Higher loss probability (p) -> lower throughput
✓ Higher RTT -> lower throughput (latency bottleneck)
✓ MSS typically 1460 bytes for Ethernet (1500 MTU – 40 byte IP+TCP header)
✓ This is the Mathis formula — used for long-lived TCP flows

🔗 Example: TCP Throughput with p=0.001

MSS	1460 bytes = 11680 bits
RTT	100 ms = 0.1 s
p	0.001, $\sqrt{p} \approx 0.0316$
Throughput	$11680 / (0.1 \times 0.0316) = 3.7 \text{ Mbps}$

🎯 13. IP Fragmentation

When a datagram is larger than the MTU, it gets fragmented into smaller pieces.

Fragments	$\text{Num fragments} = \text{ceil}((\text{Total data size}) / (\text{MTU} - \text{IP header}))$
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Offset	$\text{Offset of fragment } k = (k-1) \times (\text{data per fragment}) / 8$
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🔗 Example: Fragment 4000-byte datagram (MTU=1500, IP header=20 bytes)

Data per fragment	$1500 - 20 = 1480 \text{ bytes}$
Number of fragments	$\text{ceil}((4000 - 20) / 1480) = \text{ceil}(3980/1480) = 3 \text{ fragments}$
Fragment 1	bytes 0–1479, offset=0, MF=1
Fragment 2	bytes 1480–2959, offset=1480/8=185, MF=1
Fragment 3	bytes 2960–3979, offset=2960/8=370, MF=0 (last!)

✓ Offset field is in units of 8 bytes (always divisible by 8)
✓ MF (More Fragments) bit = 1 for all fragments EXCEPT the last
✓ DF (Don't Fragment) bit = 1 -> drop and send ICMP error if fragmentation needed
✓ Fragment offset = byte position of first byte / 8

⚠ Common Traps / Tricks

✗ Offset is in 8-byte units — divide byte offset by 8!
✗ Last fragment has MF = 0, all others MF = 1

✗ IP header is NOT fragmented — each fragment has its own 20-byte header

🎯 14. Queuing Theory — M/M/1 Queue

M/M/1: Poisson arrivals (M), Exponential service (M), 1 server.

Metric	Formula	Meaning
Utilization (ρ)	$\rho = \lambda / \mu$	Must be < 1 for stable queue
Avg queue length (L)	$L = \rho / (1 - \rho)$	Avg packets in system
Avg wait time (W)	$W = 1 / (\mu - \lambda)$	By Little's Law: $L = \lambda W$
Avg queue time (Wq)	$Wq = \rho / (\mu - \lambda)$	Wait in queue only

🔗 Example: M/M/1: $\lambda = 4$ packets/s, $\mu = 6$ packets/s

$\rho = \lambda/\mu$	$4/6 = 0.667 (< 1, \text{stable } \checkmark)$
Avg system length	$L = 0.667 / (1 - 0.667) = 2 \text{ packets}$
Avg wait in system	$W = 1 / (6 - 4) = 0.5 \text{ s} = 500 \text{ ms}$

⚠ Common Traps / Tricks

- ✗ If $\rho \geq 1$, queue grows unboundedly — system is unstable!
- ✗ λ = arrival rate (packets/s), μ = service rate (packets/s) — don't swap them
- ✗ Little's Law: $L = \lambda W$ applies to any stable system (not just M/M/1)

. Quick Formula Reference Sheet

Topic	Formula / Key Value
Transmission Delay	$T_t = L / B$
Propagation Delay	$T_p = d / v$
BDP	$BDP = B \times T_p$ (bits in flight)
a ratio	$a = T_p / T_t$
S&W Efficiency	$\eta = 1 / (1 + 2a)$
GBN Window Max	$W = 2^n - 1$
SR Window Max	$W = 2^{n-1}$
GBN/SR Efficiency	$\eta = W / (1 + 2a)$ if $W < 1+2a$, else 1
Usable Hosts	$2^h - 2$ (h = host bits)
Min Frame (Ethernet)	$2 \times T_p \times B \geq 512 \text{ bits (64 bytes)}$
Pure ALOHA	$S = G \cdot e^{-2G}$ max $\approx 18.4\%$
Slotted ALOHA	$S = G \cdot e^{-G}$ max $\approx 36.8\%$
TCP Throughput	$\approx MSS / (RTT \times \sqrt{p})$
Hamming parity bits	$2^r \geq m + r + 1$
M/M/1 Utilization	$\rho = \lambda / \mu$ (must be < 1)
M/M/1 Avg length	$L = \rho / (1 - \rho)$
IP Fragment offset	$\text{Offset} = \text{byte_pos} / 8$
Store & Forward	$\text{Total} = (n+1) \times T_t + n \times T_p$



All the best for GATE! You've got this! 